ESAD ABDULLAH KOSEDAG

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Education

Fırat University - Elazig, TURKIYE **Expected June 2028**

B.S. Software Engineering

- **GPA:** 3.25
- **Relevant Courses:**Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Discrete Mathematics, Data Science, Numerical Analysis

Professional Experience

BB Technologies - Kayseri, TURKIYE June 2023 – September 2023

Junior Software Developer

- Co-developed a Gantt-chart-based production scheduling system for a steel manufacturing plant, covering six machine types, and integrated it with the company's existing .NET infrastructure using Node.js.
- Built the system to schedule machine operations down to individual worker shift synchronization, enabling significantly more precise production planning than the company's previous manual process.
- Per finance team estimates, the tool contributed to **an estimated $1–4M in annual cost savings** and accelerated production planning timelines by **at least one month**.
- Used Devextreme for custom UI components and managed data across FirebirdSQL and MSSQL databases.

Projects and Outside Experience

OncoMind - GitHub - Elazig, TURKIYE October 2025 – Present

Project Founder — AI/ML Research (LSTM, Graph-VAE Model Training and Scientific research)

- Curated and preprocessed a dataset of **6 kinase inhibitors**, cleaning 57,619 raw molecular entries down to **32,437 canonicalized, deduplicated compounds** for generative model training.
- Trained and benchmarked two generative architectures for de novo drug candidate synthesis (SMILES): an LSTM model and a Graph-VAE model, achieving **95% and 90% accuracy** respectively in generating chemically viable compounds.
- Identified that LSTM-generated candidates exhibited **90% BBB toxicity**; re-engineered the pipeline with the Graph-VAE architecture to **reduce BBB toxicity by 10%**, improving candidate safety profile for targeted therapy.
- Architected a full-stack drug discovery platform (Python/FastAPI backend, C#/.NET service layer, React frontend, MongoDB) to support the end-to-end pipeline from data ingestion to candidate generation.

Alcin UAV Team - Kayseri, TURKIYE January 2023 – June 2024

Computer Vision & UAV Systems Engineer

- Optimized and deployed a custom YOLOv8 model on an NVIDIA Jetson Xavier NX, achieving stable 30 FPS inference with **97% detection accuracy** for real-time UAV tracking.
- Achieved uninterrupted target lock for **60 seconds in stable flight conditions** and **25 seconds in turbulent conditions**, validating tracking robustness under variable aerodynamic disturbance.
- Designed and developed flight control algorithms and a Ground Control Station (GCS) using Qt Quick for real-time UAV avionics monitoring.

Capella Model Rocket Team - Kayseri, TURKIYE November 2022 – June 2024

Team Leader, Lead Avionics & Embedded Systems

- Programmed a competition-mandated telemetry system on an **STM32F103C8T6(LQFP48)** flight computer, compressing sensor data packets to **78 bytes** to meet judge interface bandwidth requirements while preserving full telemetry fidelity.
- Implemented a **Kalman filter** for sensor fusion, reducing telemetry data loss from **~30% to 5–8%** under flight conditions, significantly improving ground station tracking reliability.
- Designed and developed a Ground Control Station (GCS) interface using C# Windows Forms to visualize real-time flight data.

BudgetTravel - GitHub - Remote *February 2025 – June 2025*

Scrum Master & AI Engineer (LLM Training + Agent Integration)

- Reduced AI-generated travel itinerary planning time from **~10 minutes to 2–5 minutes** by optimizing the agent's background review and map-traversal pipeline, integrating OpenAI and Google Maps APIs for real-time route optimization.
- Led sprint planning and task delegation as Scrum Master for a 4-person capstone team, coordinating full-stack development across AI backend and web frontend workstreams.
- Designed a full-stack AI-powered travel planning platform enabling users to generate personalized, optimized itineraries from natural language input.

Skills and Interests

Skills: Python, C++, Docker, Team Leadership, Machine Learning, Rdkit, Node.js, JavaScript, PyTorch, TensorFlow, Scikit-Learn, OpenCV, Pandas, Microsoft Office, Adobe Illustrator.

Languages: Turkish (native), English (intermediate/advanced).

Interests: Film Production, Game Production, Composing Music, Graphic Design.